

5. Probabilistic models 1

Machine learning

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Probabilistic models - lectures outline

- Probability introduction + Bayes rule
- Probabilistic models - parameters estimation:
 - Maximum likelihood
 - Maximum a-posteriori
 - Bayesian inference
- Probabilistic models - example applications:
 - Naive Bayes classifier
 - Gaussian mixture model – EM algorithm
 - Logistic regression

Probabilistic models 1

Probabilistic models 2

Probability introduction

Definitions

Probability – is a measure quantifying the likelihood that events will occur. The probability of an event A is written as:

$$P(A) \text{ (or } p(A) \text{ or } \Pr(A))$$

Random variable – is a variable whose value is uncertain
– e.g. dice roll, covid test result, class label in dataset

$$X$$

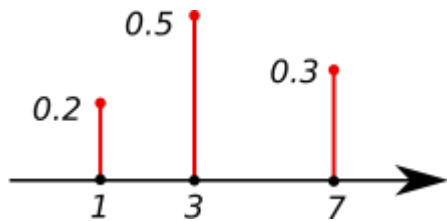
Probability of random variable X to have value x_1 : $P(X = x_1)$

Probability distribution - provides the probabilities of occurrence of different possible outcomes of random variable (values $P(X = x_i)$ for all x_i)

Probability distribution

Probability mass function (PMF)

$$P(X = x) = p(x)$$

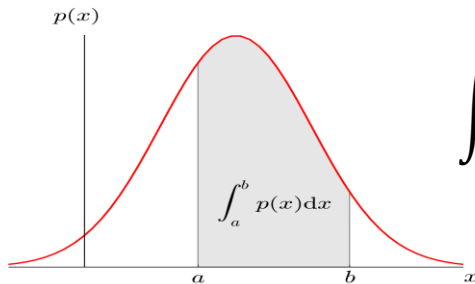


$$\sum_x p(x) = 1$$

Probability density function (PDF)

$$P(X = x) = 0$$

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} p(x) dx$$



$$\int_{-\infty}^{\infty} p(x) dx = 1$$

$$p(x) = p_X(x) = P(X = x) \quad \text{or} \quad p(x) \text{ for PDF}$$

Probability distribution – non-parametric

Non-parametric description – only for a **discrete** distribution and it is described directly by probabilities of each discrete variable value.

Examples

$$x \in \{1, 3, 7\}$$

$$P(X = 1) = 0.1$$

$$P(X = 3) = 0.4$$

$$P(X = 7) = 0.5$$

$$x \in \{0.1, 1.5, 2.3, 26.7\}$$

$$P(X = 0.1) = 0.10$$

$$P(X = 1.5) = 0.25$$

$$P(X = 2.3) = 0.38$$

$$P(X = 26.7) = 0.27$$

$$x \in \{0, 1\}$$

$$y \in \{0, 1\}$$

$$P(X = 0, Y = 0) = 0.1$$

$$P(X = 1, Y = 0) = 0.2$$

$$P(X = 0, Y = 1) = 0.4$$

$$P(X = 1, Y = 1) = 0.3$$

Probability distribution – non-parametric

Feature table

<i>Subject</i>	<i>Feature 1</i>	<i>Decision</i>
01	Blue	Positive
02	Blue	Negative
03	Brown	Positive
...	...	...
N	Green	Positive

Contingency table

	<i>Blue</i>	<i>Green</i>	<i>Brown</i>	Total
Positive	53	47	45	145
Negative	65	33	82	180
Total	118	80	127	325

Probability distribution (divide by sum)

	<i>Blue</i>	<i>Green</i>	<i>Brown</i>
Positive	0.16	0.15	0.14
Negative	0.20	0.10	0.25

From observation of discrete variables (Feature table), we can derive any Contingency table, but not the other way around.

What happened if we have 3 features?

Can I do contingency table from a continuous random variable?

Probability distribution – parametric

Parametric description - The shape of probability distribution is defined by mathematical description with several parameters. It can be used for both continuous or discrete variables.

E.g. 1D normal distribution is defined by a mean value μ and a standard deviation σ

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Generally, we have a vector θ of parameters, e.g. for 1D normal distribution $\theta = [\mu, \sigma]$.

Example for a Bernoulli distribution

Parametric

$$f(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

$$x \in \{0, 1\}$$

$$\theta = 0.4$$

Feature table

<i>Subject</i>	<i>Feature 1</i>
01	1
02	0
03	0
...	...
N	1

Non-parametric

$$P(X = 0) = 0.6$$

$$P(X = 1) = 0.4$$

If we have available observed data of a discrete random variable, we can obtain both parametric or non-parametric probability distribution. But, for continuous variables, there is only parametric description.

Joint probability

Joint probability - is the probability of two events (X,Y) happening together.

Joint probability distribution - is the joint probability function of two or more random variables.

Discrete case (mass function)

$$P(X = x, Y = y) = P(X = x \text{ and } Y = y) = p(x, y)$$

Continuous case (density function)

$$P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2); p(x, y)$$

For independent variables

$$p(x, y) = p(x) \cdot p(y)$$

The following applies to a joint probability distribution:

$$p(x, y) = p(y, x)$$

For dependent variables – chain rule of probability

$$p(x, y) = p(x|y) \cdot p(y)$$

Marginal probability

– is defined as a sum of a joint probability function of two or more random variables along specific dimensions. Thus, some variables are discarded (so called *marginalized out*) and this process is named *marginalization*.

Marginalizing along Y random variable

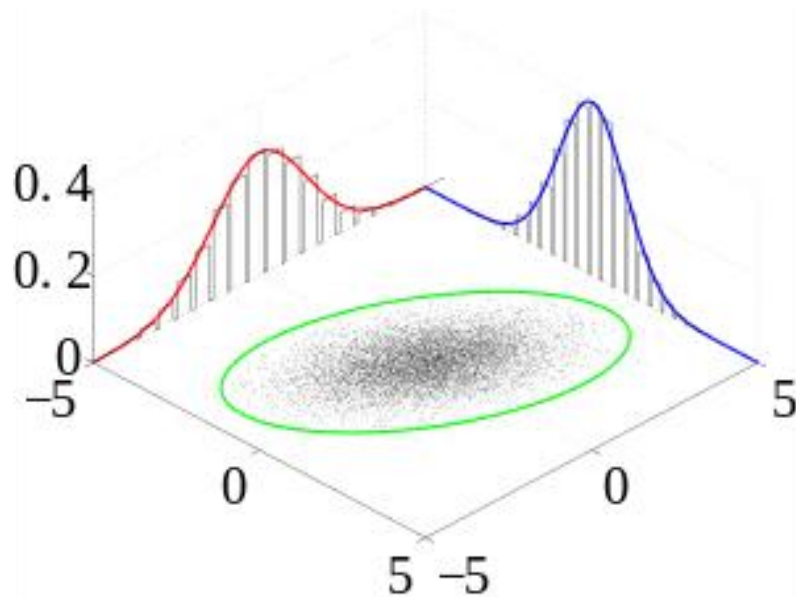
$$P(X = x) = \sum_y P(X = x, Y = y)$$

Marginalizing along X random variable

$$P(Y = y) = \sum_x P(X = x, Y = y)$$

Abbreviated notation – Law of total probability

$$p(x) = \sum_y p(x, y)$$



Conditional probability

Conditional probability – is a value of probability, that an event B will occur if another event A has occurred. The concept of conditional probability is one of the most fundamental and one of the most important in probability theory.

$$P(X|Y = y)$$

Kolmogorov definition: conditional probability of events A and B is given a rate of probability that both events A and B occur and marginal probability of event B.

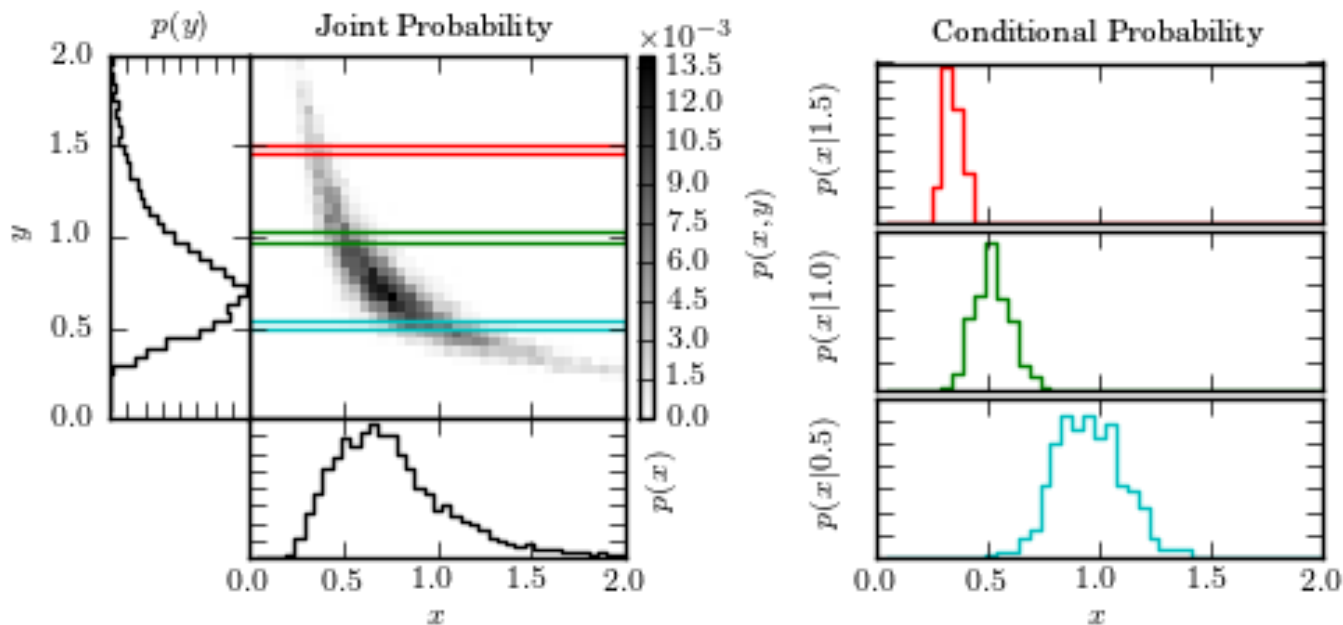
$$p(x|y) = \frac{p(x, y)}{p(y)}$$

$$p(x, y) = p(x|y) \cdot p(y)$$

In contrast with joint probability, the conditional probability of event A given B is not equal to the probability of event B given A.

$$p(x|y) \neq p(y|x)$$

Joint/conditional/marginal probability example



Probability distribution example

X – shape – c (circle), s (square)

Y – color – b (blue), o (orange)

$$P(X = c) = p(c)$$

$$P(X = s) = p(s)$$

$$P(Y = b) = p(b)$$

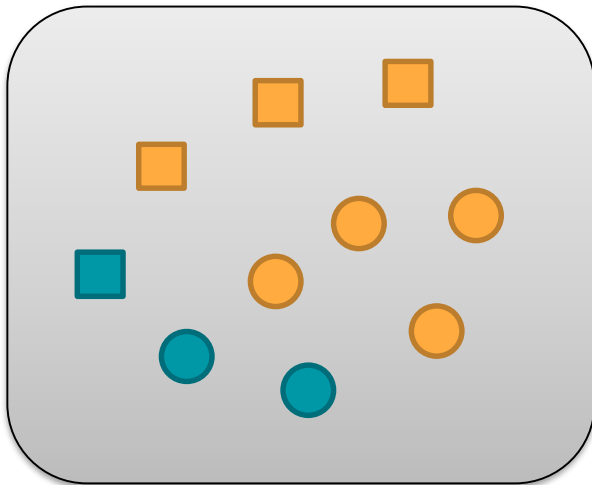
$$P(Y = o) = p(o)$$

$$p(b, s)$$

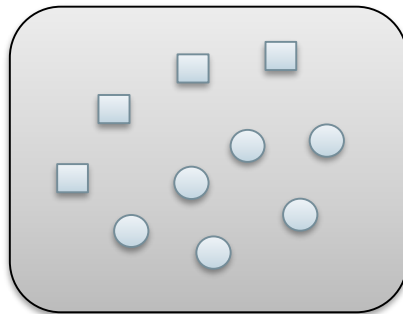
$$p(b|s)$$

$$p(b|s) \neq p(s|b)$$

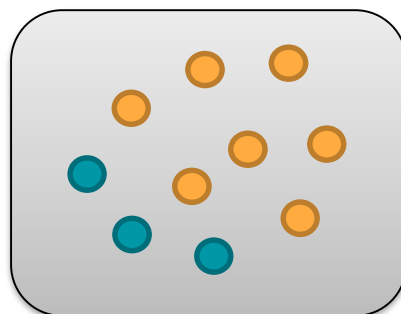
$$p(b, s) = p(s, b)$$



X – shape



Y – color



$$p(x, y) = p(x|y) \cdot p(y)$$

$$p(x|y) = \frac{p(x, y)}{p(y)}$$

$$p(y) = \sum_x p(x, y)$$

$$p(y) = \sum_{i=1}^n p(y|x_i)p(x_i)$$

Bayes' rule

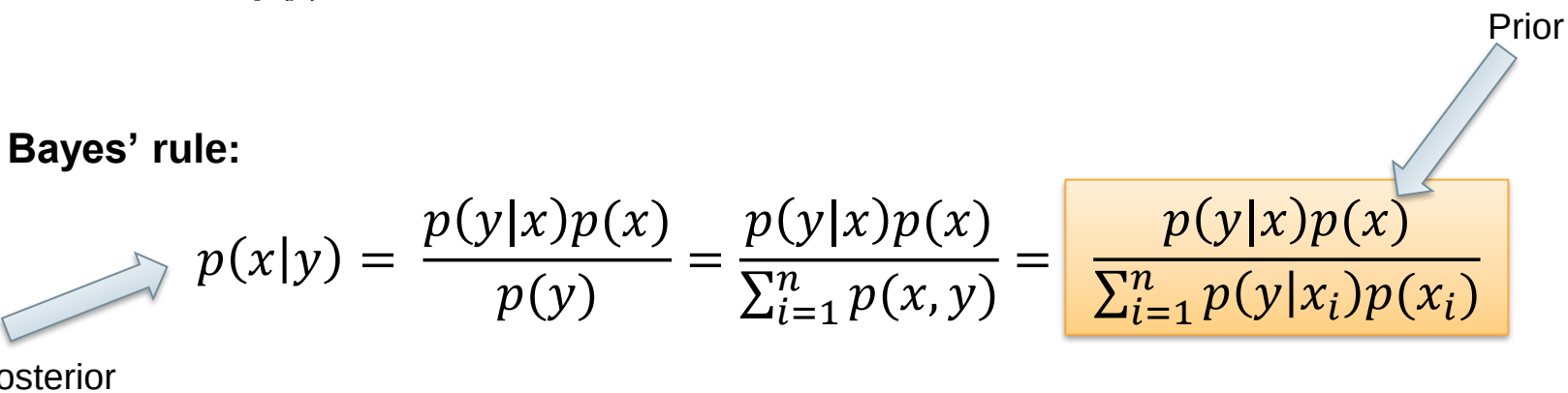
From definition of conditional probability:

$$p(x, y) = p(x|y)p(y) = p(y, x) = p(y|x)p(x)$$

$$p(x|y)p(y) = p(y|x)p(x)$$

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

Bayes' rule:


$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} = \frac{p(y|x)p(x)}{\sum_{i=1}^n p(x, y)} = \frac{p(y|x)p(x)}{\sum_{i=1}^n p(y|x_i)p(x_i)}$$

Posterior

Prior

Bayes rule example 1

A ...a person has a heart attack

B₁ ...a person is non – smoker

B₂ ...a person is light smoker

B₃ ...a person is heavy smoker

Probability:

$$p(B_1|A) = 0,5$$

$$p(B_3|A) = 0,2$$

Prior:

$$p(B_1) = 0,9$$

$$p(B_2) = 0,07$$

$$p(B_3) = 0,03$$

$$p(A) = 0,05$$

Measurement:

	<i>B₁</i>	<i>B₂</i>	<i>B₃</i>	Total
<i>A</i>	50	30	20	100

Bayes rule example 1

A ...a person has a heart attack

B₁ ...a person is non – smoker

B₂ ...a person is light smoker

B₃ ...a person is heavy smoker

Measurement:

	<i>B₁</i>	<i>B₂</i>	<i>B₃</i>	Total
A	50	30	20	100

Probability:

$$p(B_1|A) = 0,5$$

$$p(B_2|A) = 0,3$$

$$p(B_3|A) = 0,2$$

Prior:

$$p(B_1) = 0,9$$

$$p(B_2) = 0,07$$

$$p(B_3) = 0,03$$

$$p(A) = 0,05$$

Posteprior:

$$p(A|B_1) = ???$$

$$p(A|B_3) = ???$$

$$p(A|B_1) = \frac{p(B_1|A)p(A)}{p(B_1)} = \frac{0,5 \cdot 0,05}{0,9} = 0.0278$$

$$p(A|B_3) = \frac{p(B_3|A)p(A)}{p(B_3)} = \frac{0,2 \cdot 0,05}{0,03} = 0,33$$

Bayes rule example 2

A ... disease occurs

B₊ ... positive result of test with sensitivity and specificity of 98%

Case 1: If I have a positive test. What probability is that I really have a disease?

by Bayes' probability:

we have available a new findings about prevalence of disease – 10%

$$p(A|B_+) = \frac{p(B_+|A)p(A)}{p(B_+)} = \frac{0,98 \cdot 0,1}{0,116} = 84\%$$

$$p(B_+) = p(B_+, A) + p(B_+, \bar{A}) = p(B_+|A)p(A) + p(B_+|\bar{A})p(\bar{A}) = 0,98 \cdot 0,1 + 0,02 \cdot 0,9 = 0,116$$

$$p(y) = \sum_{i=1}^n p(y|x_i)p(x_i)$$

Bayes rule example 2

A ... disease occurs

B₊ ... positive result of test with sensitivity and specificity of 98%

Case 1: If I have a positive test. What probability is that I really have a disease?

by Bayes' probability:

we have available a new findings about prevalence of disease – 1%

$$p(A|B_+) = \frac{p(B_+|A)p(A)}{p(B_+)} = \frac{0,98 \cdot 0,01}{0.0296} = 34\%$$

$$p(B_+) = p(B_+, A) + p(B_+, \bar{A}) = p(B_+|A)p(A) + p(B_+|\bar{A})p(\bar{A}) = 0,98 \cdot 0,01 + 0,02 \cdot 0,99 = 0.0296$$

Model parameters estimation

Probabilistic models incorporate random variables and probability distributions into the model of an event or phenomenon. While a deterministic model gives a single possible outcome for an event, a **probabilistic model gives a probability distribution as a solution.**

Generative vs Discriminative models

Discriminative



$$f: X \rightarrow Y$$

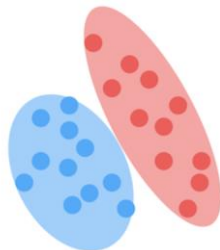
SVM

k-NN

$$p(Y|X = x)$$

Logistic regression

Generative



$$p(X, Y)$$

Naive Bayes classifier

Generative Adversarial
Networks (GANs)

Generative without classes



$$p(X)$$

Gaussian (fitted with
MLE or MAP)

GANs

Model parameters probability distribution

We want model of probability distribution $p(x)$.

It can be specified by some parametric description with parameters θ : $p_{\theta}(x)$

Model parameters have again some probability distribution given measured dataset D : $p(\theta|D)$

$(p(\theta|x))$ will be used in this presentation)

If we know $p(\theta|D)$, we can select most probable model parameters and use them for in $p_{\theta}(x)$ model.

$$p(x|y) = \frac{p(y|x) \cdot p(x)}{p(y)}$$

Bayes' terminology for model estimation

Prior probability $p(\theta)$

is the distribution of the parameters before any data is observed (what we know before performing the experiment).

Likelihood $p(x|\theta)$

is the distribution of the observed data conditional on its parameters.
It quantifies the likelihood that model with parameters θ will generate data x .

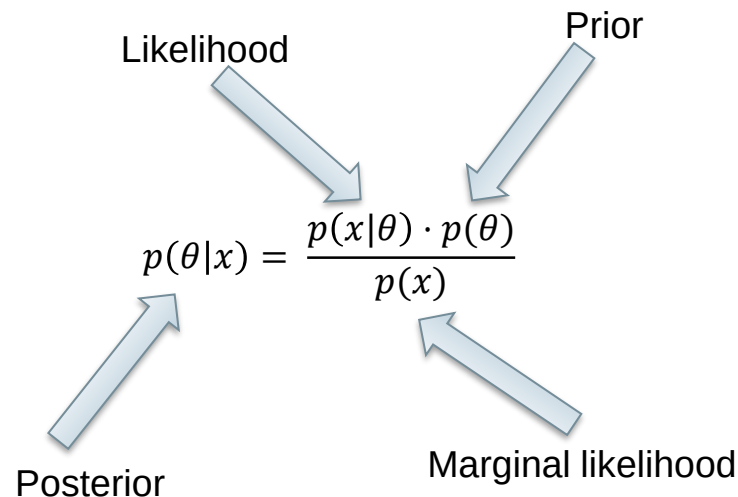
Posterior probability $p(\theta|x)$

is the distribution of the parameters after taking into account the observed data. It is the probability of model parameters given measured data.

Marginal likelihood $p(x) = \sum_{i=1}^n p(x|\theta_i)p(\theta_i)$

(evidence) is the distribution of the observed data marginalized over the parameters.

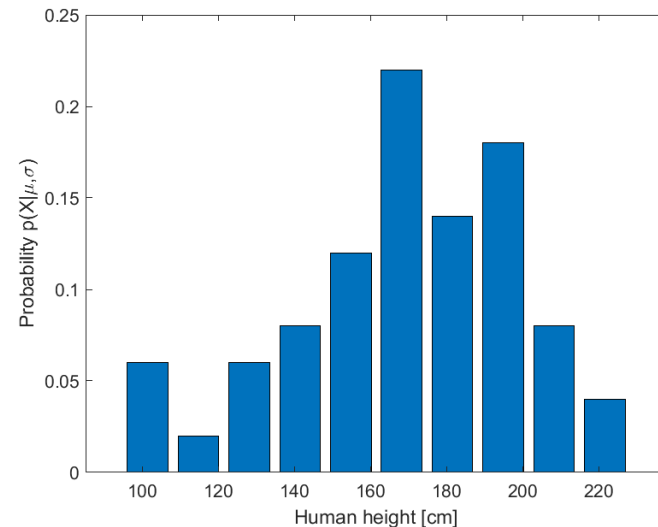
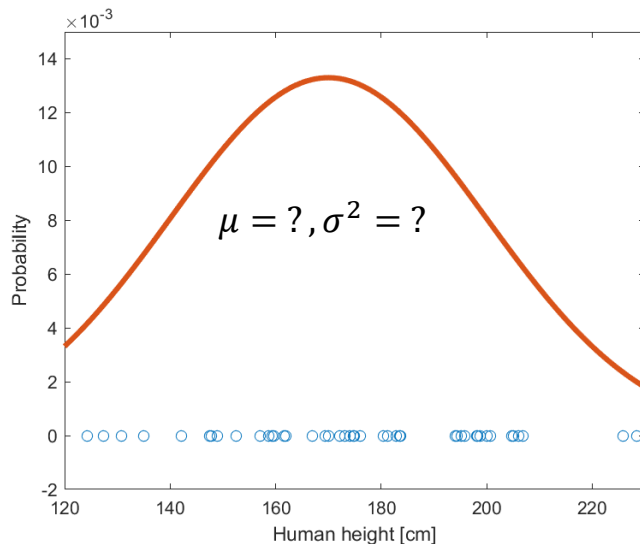
Data x and model parameters θ



Example – model parameters estimation

We have N measurements of human height $X = x_1, x_2, \dots, x_N \sim \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$

$x_1 = 172 \text{ cm}$
 $x_2 = 189 \text{ cm}$
 $x_3 = 168 \text{ cm}$
 $\vdots$
 $x_N = 163 \text{ cm}$

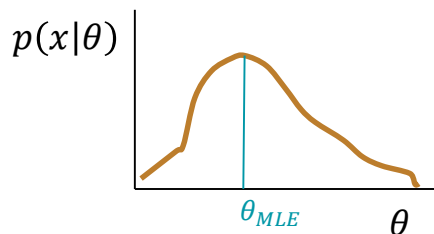


We are interested in model parameters probability distribution $p(\theta|x)$
where we want found the most probable parameters $\operatorname{argmax}_{\theta} p(\theta|x)$

Parameter estimation methods

$$p(\theta|x) = \frac{p(x|\theta) \cdot p(\theta)}{p(x)}$$

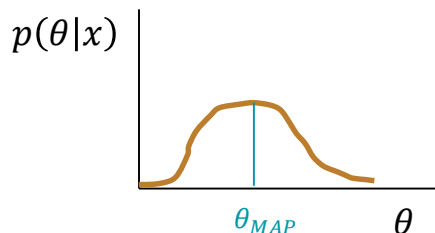
Maximum likelihood estimation
MLE



Under the assumed statistical model the observed data are most probable.

$$\operatorname{argmax}_{\theta} p(x|\theta)$$

Maximum a-posteriori probability
MAP



Find most probable model parameters for given data.

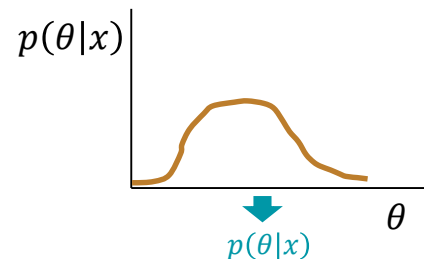
$$\operatorname{argmax}_{\theta} p(\theta|x) =$$

$$= \operatorname{argmax}_{\theta} \frac{p(x|\theta) p(\theta)}{p(x)} =$$

$$= \operatorname{argmax}_{\theta} p(x|\theta) p(\theta)$$

constant
can be
neglected for
argmax

Bayesian inference



Find model parameters probability distribution for given data.

$$p(\theta|x) =$$

$$= \frac{p(x|\theta) p(\theta)}{p(x)} =$$

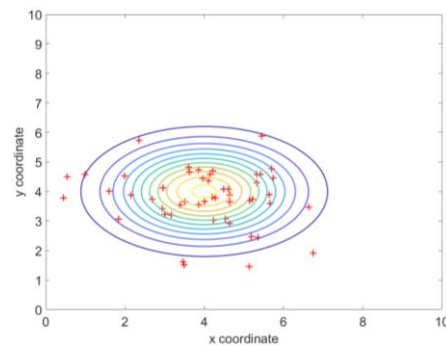
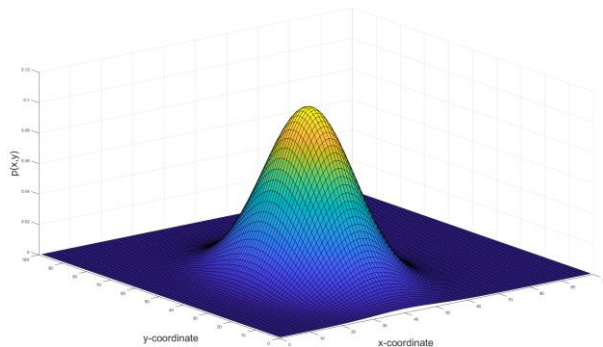
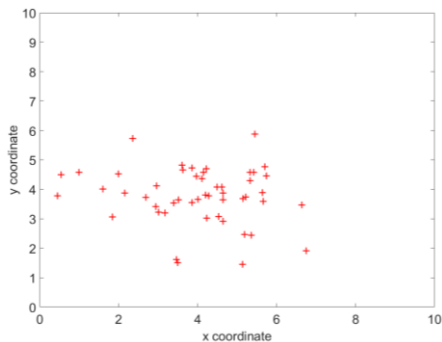
$$= \frac{p(x|\theta) p(\theta)}{\int_{\theta} p(x|\theta) p(\theta) d\theta}$$

hard to
calculate

Maximum likelihood

Maximum likelihood (ML, MLE)

The estimation method that finds the parameters of a probability distribution based on maximizing a **likelihood** function for the set of observed data $x = \{x_1, x_2, \dots, x_n\}$. Thus, the task is to find argument of maxima of likelihood function. The resulting model **is the most likely model for generating the observed data**.



The main advantage is an analytic solution to many problems and thus easy to find the parameters of the preselected distribution. Nevertheless, it can be solved also any optimization technique (brute-force, iterative algorithm).

Definition of ML

Task:

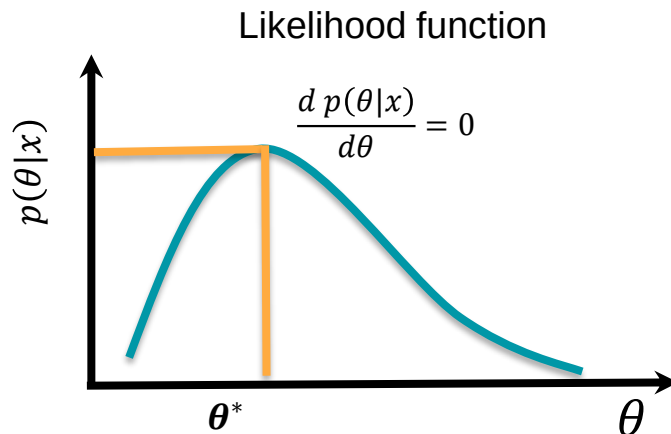
to find the θ that maximizes likelihood:

$$\theta^* = \operatorname{argmax}_{\theta} p(x|\theta)$$

if we know that $x_1, x_2, \dots, x_n \sim \mathcal{N}(\theta)$, where $\theta = [\mu, \sigma^2]$

$$\mu^* = \operatorname{argmax}_{\mu} p(x|\mu, \sigma^2)$$

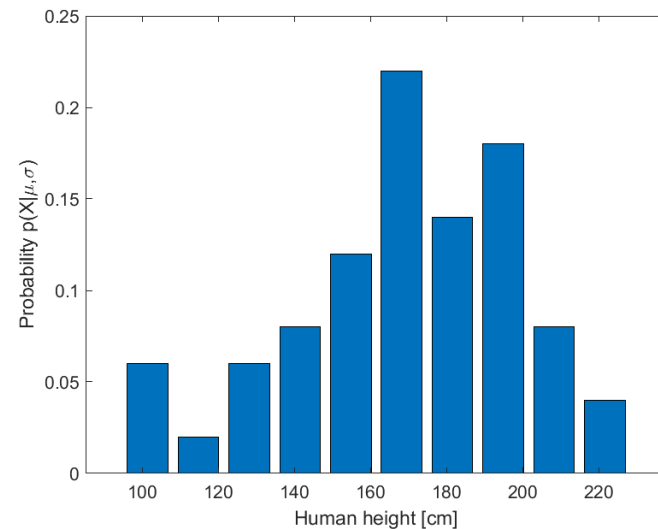
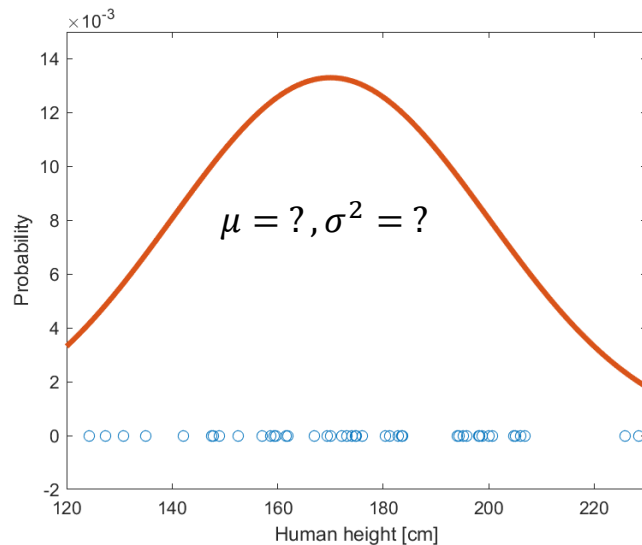
$$\sigma^{2*} = \operatorname{argmax}_{\sigma^2} p(x|\mu, \sigma^2)$$



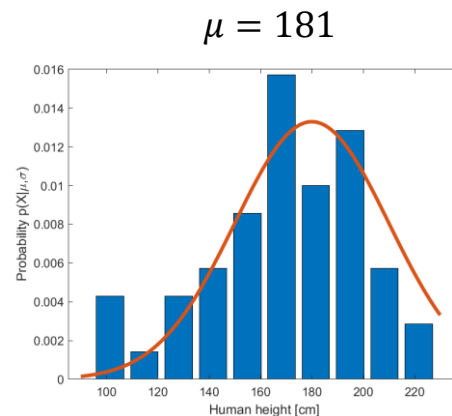
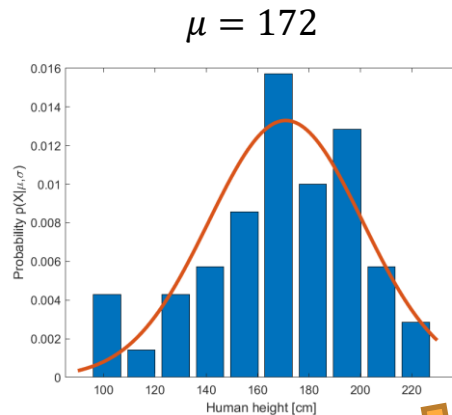
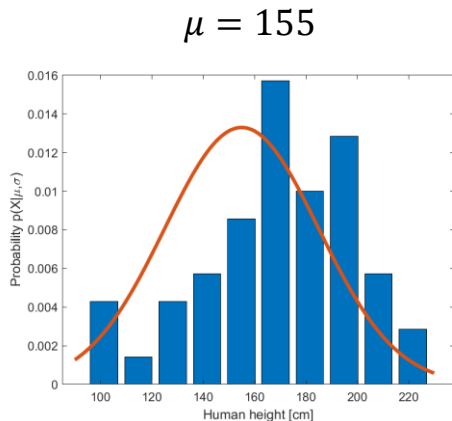
Example

We have N measurements of human height $X = x_1, x_2, \dots, x_N \sim \mathcal{N}(\mu, \sigma^2)$

$x_1 = 172 \text{ cm}$
 $x_2 = 189 \text{ cm}$
 $x_3 = 168 \text{ cm}$
 $\vdots$
 $x_N = 163 \text{ cm}$



Likelihood function



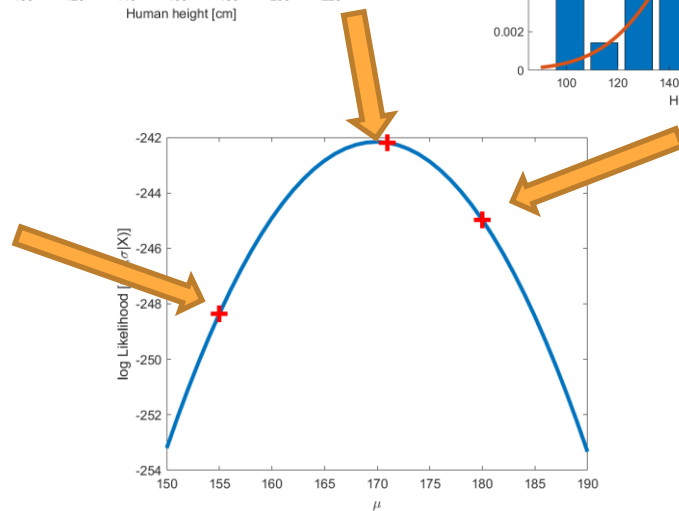
PDF of normal distribution,

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

for n **independent identically distributed (i.i.d.)** random samples
(the likelihood):

$$x_i \sim \mathcal{N}(\mu, \sigma^2) \text{ i.i.d.}$$

$$p(x_1, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n p(x_i, | \mu, \sigma^2) = \prod_{i=1}^n p(x_i, | \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$



Solution – for Normal PD

PDF of normal distribution,

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

for n **independent identically distributed (i.i.d.)** random samples (the likelihood):

$$x_i \sim \mathcal{N}(\mu, \sigma^2) \text{ i.i.d.}$$

$$p(x_1, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n p(x_i | \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\log p(\mathbf{x}|\boldsymbol{\theta}) = \log p(x_1 \dots, x_n | \mu, \sigma^2)$$

Solution:

$$\mu^* = \operatorname{argmax}_{\mu} \mathcal{L}(\mu, \sigma^2) = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

$$\sigma^{2*} = \operatorname{argmax}_{\sigma^2} L(\mu, \sigma^2) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \hat{\sigma}^2$$

Can be proved:

$$E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$$

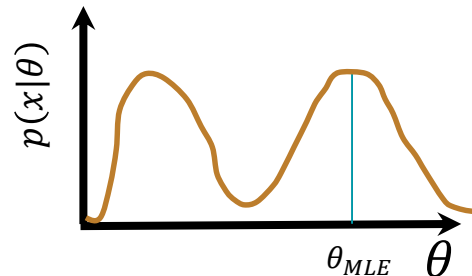
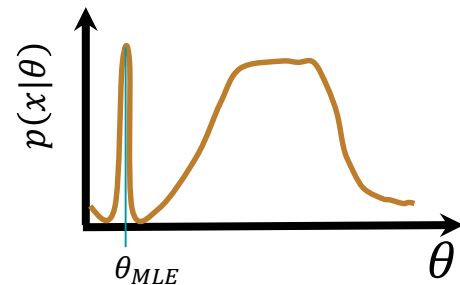
MLE pros and cons

Pros:

- Easy to compute (closed form solution or simple maximization problem)
- Intuitive interpretation (e.g. mean for Normal or Bernoulli distributions)
- Good asymptotic properties – unbiased, efficient, consistent

Cons:

- Point estimate – no representation of uncertainty
- Not suitable for the multi-modal likelihood
- Existence & uniqueness of ML is not guaranteed
- Maybe not what we want.
- Overfitting (e.g. „black swan paradox“)

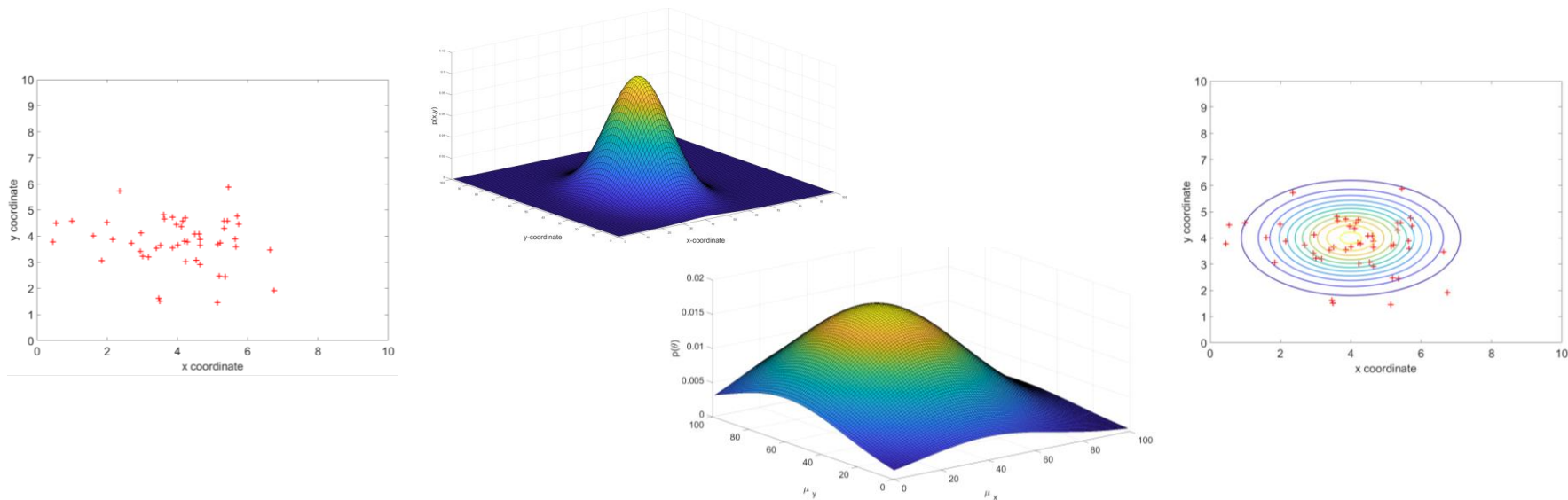


Maximum a-posteriori probability

Maximum a-posteriori probability (MAP)

The estimation method that find the parameters of a probability distribution based on maximizing a **a-posterior probability** $p(\theta|x)$ for the set of observed data $x = \{x_1, x_2, \dots, x_n\}$.

The resulting model, which is defined by estimated parameters, **is the most probable model for given (observed) data** x taking into account a prior distribution of individual models.



Definition

Maximum Likelihood

$$\theta_{MLE}^* = \operatorname{argmax}_{\theta} p(x|\theta)$$

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{\cancel{p(x)}}$$

we assume a
uniform distribution
of probability

for finding the
minimum of function
is irrelevant

Maximum a-posteriori probability

$$\theta_{MAP}^* = \operatorname{argmax}_X p(\theta|x) = \operatorname{argmax}_{\theta} \frac{p(x|\theta) \cdot p(\theta)}{\cancel{p(x)}}$$

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta} p(x_1, \dots, x_n|\theta) \cdot p(\theta)$$

- regularization of ML estimation
- using a prior distribution of a related event
- the posterior distribution $p(\theta|x)$ can be calculated using Bayes' theorem from likelihood function $p(x|\theta)$ and the prior $p(\theta)$

Conjugate prior

If the posterior distribution $p(\theta|x)$ and the prior $p(\theta)$, are from the same probability distribution family (only exponential) then they are called conjugate distributions and the prior is called conjugate prior for the likelihood function $p(x|\theta)$.

$$p(\theta|x) \propto p(x|\theta) \cdot p(\theta)$$

If we want to ensure the same distribution of the posterior $p(\theta|x)$ and a prior $p(\theta)$, we need to choose an appropriate conjugate distributions for a likelihood distribution $p(x|\theta)$.

Examples:

- normal distribution of the likelihood function needs normal distribution for a prior,
- Bernoulli distribution needs beta distribution,
- more at Table of conjugate distributions: https://en.wikipedia.org/wiki/Conjugate_prior

WHY?

- analytic solution of the product $p(x|\theta)$ and $p(\theta)$ will be less difficult than with another distribution
- more transparent expression of a prior distribution update (by likelihood function)

Solution – for Normal PD

PDF of Normal distribution

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

likelihood function PDF for n **independent** random samples

$$p(x_1, \dots, x_n|\mu, \sigma^2) = \prod_{i=1}^n p(x_i|\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}\right) \quad x_i \sim \mathcal{N}(\mu, \sigma^2) \text{ i.i.d.}$$

prior PDF according to **conjugate prior** for Normal distribution with known σ of likelihood fcn.

$$p(\mu|\mu_P, \sigma_P^2) = \frac{1}{\sqrt{2\pi\sigma_P^2}} \cdot \exp\left(-\frac{(\mu - \mu_P)^2}{2\sigma_P^2}\right) \quad \mu \sim \mathcal{N}(\mu_P, \sigma_P^2)$$

$\mu_P, \sigma_P^2, \sigma^2 - \text{constants}$

$$\mathcal{P}(\mu; \sigma^2, \mu_P, \sigma_P^2) = \log [p(x_1 \dots, x_n, |\mu, \sigma^2) \cdot p(\mu|\mu_P, \sigma_P^2)]$$

Solution:

$$\mu_{MAP}^* = \operatorname{argmax}_{\mu} \mathcal{P}(\mu; \sigma^2, \mu_P, \sigma_P^2) = \frac{\overset{\text{weights}}{\underset{\text{sample mean (ML)}}{n\sigma_P^2}} \cdot \overset{\text{convex combination of } \bar{y} \text{ and } \mu}{\bar{x}} + \frac{\sigma^2}{n\sigma_P^2 + \sigma^2} \cdot \overset{\text{prior mean}}{\mu_P}$$

Summary

Pros:

- Analytic solution is usually possible when conjugate priors are used (for exponential family distributions only)
 - otherwise numerical optimization can be used (gradient methods, Newton's method, Monte Carlo simulation, EM algorithm)
- Easy to compute (but harder than MLE), interpretable
- Avoid overfitting – regularization
- Converges to MLE with enough data

Cons:

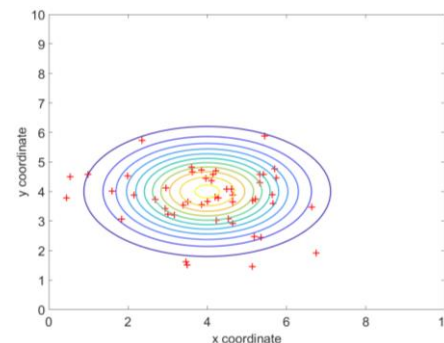
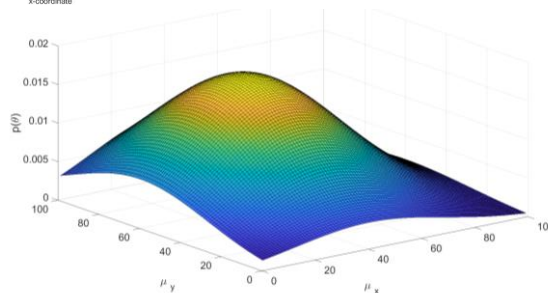
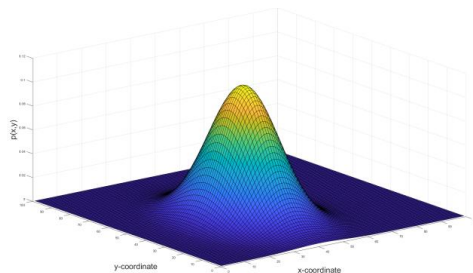
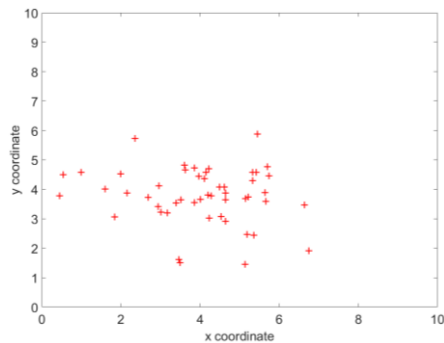
- Point estimate – no representation of uncertainty
- Must assume a prior on θ

Bayesian inference

Bayesian inference

The **estimation method** that find the **probability distribution of model parameters** for the set of observed data $x = \{x_1, x_2, \dots, x_n\}$.

The resulting distribution of models, taking into account a prior distribution of individual models.



Bayesian inference

- We do not get some estimate of θ^* but whole probability distribution of $p(\theta|x)$

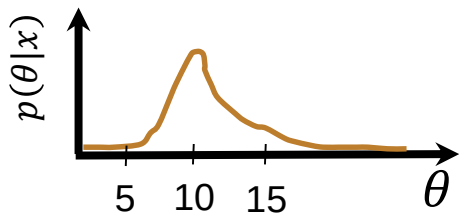
$$p(\theta|x) = \frac{p(x|\theta) p(\theta)}{p(x)} = \frac{p(x|\theta) p(\theta)}{\int_{\Theta} p(x|\theta) p(\theta) d\theta}$$

Hard to calculate integral, which cannot be neglected without argmax.



Approximation methods - Markov Chain Monte Carlo (MCMC) algorithms (Gibbs sampling, Metropolis-Hastings), Variational Inference

- Closed form solution for conjugate priors. - very complicated – we will not use them.
- With known distribution, we have uncertainty for estimated parameters:



We know not only that best model is $\theta = 10$, but we can calculate that $5 < \theta < 15$ with probability 92% ($p(5 < \theta < 15|x) = 0.92$)

Posterior predictive distribution

- For MLE and MAP we get single model with optimal θ^* and output distribution is then defined using single model:

$$p(x_{predict}|\theta^*) = p_{\theta^*}(x_{predict})$$

- For Bayesian inference (know posterior distribution of θ), we can calculate posterior predictive distribution, which consider all models with their probability:

$$p(x_{predict}|x) = \int p(x_{predict}|\theta) p(\theta|x) d\theta$$

- It still have closed form solution for conjugate prior, and it is more precise prediction

Bayesian inference with conjugate priors - examples

Likelihood – Bernoulli, Prior – Beta, Posterior – Beta:

$$\text{Beta}(\alpha', \beta') = \text{Bernoulli}(\theta) \text{Beta}(\alpha, \beta)$$

$$p(x \mid \theta) = \theta^x (1 - \theta)^{(1-x)}$$

$$\text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{1-\alpha} (1 - \theta)^{1-\beta}$$

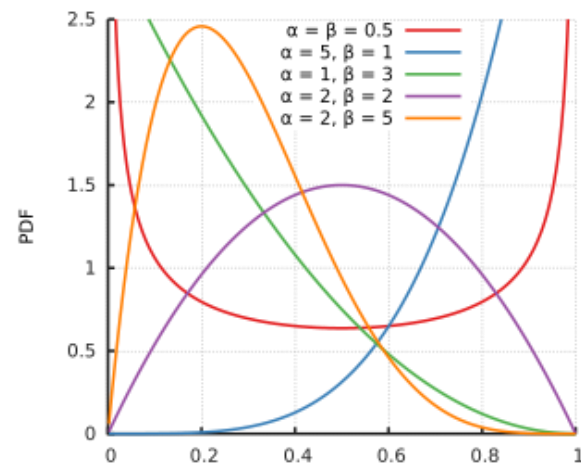
Posterior distribution parameters:

$$\alpha' = \alpha + \sum_{i=1}^n x_i$$

$$\beta' = \beta + n - \sum_{i=1}^n x_i$$

Posterior predictive:

$$p(\tilde{x} = 1) = \frac{\alpha'}{\alpha' + \beta'}$$



$$\theta_{MAP} = \frac{\sum_{i=1}^n x_i + \alpha - 1}{n + \beta + \alpha - 2}$$

Bayesian inference with conjugate priors - examples

Likelihood - Normal with known variance σ^2 , Prior – Normal, Posterior - Normal

$$Normal(\mu', \sigma') = Normal(\mu; \sigma) Normal(\mu_0, \sigma_0)$$

$$N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Posterior distribution parameters:

$$\mu' = \frac{n\sigma_p^2}{n\sigma_p^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_p^2 + \sigma^2} \cdot \mu_0 = \mu_{MAP}$$

$$\sigma^{2'} = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1}$$

$$\mu_{MLE} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Posterior predictive: $N(x|\mu', \sigma^{2'} + \sigma^2)$

Bayesian inference

Pros:

- Not point estimate – provide whole probability distribution of model parameters.
=> We know model uncertainty.
- Avoid overfitting.
- Can provide exact posterior predictive distribution.
- Automatic model selection.

Cons:

- Must assume a prior.
- Exact computation can be intractable (integral).

Summary

MLE and MAP are easy to compute and intuitively interpretable often solved using an analytical solution. Even without analytical solution, optimization is possible. However, it provides only single estimate of parameters without uncertainty.

Maximum likelihood

- MLE estimation attempts to find the parameter vector θ that maximizes $p(x|\theta)$ – likelihood, thus the model that the best generates observed data.
- The estimation is based only on the observed data x .
- Overfitting

Maximum a posteriori probability

- MAP estimation attempts to find the parameter vector θ that maximizes $p(\theta|x)$ – posterior, thus the most probable model (parameters of distribution) assuming observed data x .
- The estimation requires a prior $p(\theta)$, which favors/disadvantages of individual distributions.
- It has a regularization in the form a prior (overfitting elimination).

Bayesian inference is hard to compute (except for special cases). However, it provide full probability distribution of model parameters. We can also find posterior predictive distribution – which is mix of prediction of all models with their probability.

- Similar to MAP, but provides full distribution.

Example

MLE/MAP – blood pressure

A ... is normal distribution of measurement $\mathcal{N}(\mu, \sigma^2)$

B ... is set of measurement of blood pressure from one subject $B = \{x_1, x_2, \dots, x_n\}$

We assume a normal distribution of a blood pressure measurement.

The task is to find parameters of the posterior distribution $A - p(A, B)$

i.e. mean $\hat{\mu}$ and variance $\hat{\sigma}^2$, which will best represents measured data B .

In this example we consider known σ as a standard deviation of used sphygmomanometer – estimation is simplified to the estimation of mean μ only.

MLE Solution:

$$\mu_{ML}^* = \operatorname{argmax}_{\mu} \log p(B|\mu; \sigma^2) = \bar{x}$$

MAP Solution:

For MAP estimation we need to know a prior distribution of a related event

C – conjugate distribution with $\mathcal{N}(\mu_0, \sigma_0^2)$, with known μ_0 – mean value of blood pressure in population and σ_0^2 – standard deviation in population.

$$\begin{aligned} \mu_{MAP}^* &= \operatorname{argmax}_{\mu} \mathcal{P}(\mu; \sigma^2, \mu_0, \sigma_0) = \operatorname{argmax}_{\mu} \log [p(B|\mu, \sigma^2) \cdot p(\mu|\mu_0, \sigma_0^2)] = \\ &= \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \cdot \mu_0 \end{aligned}$$

Example

MLE / MAP – blood pressure

	x_1	x_2	x_3
Observed	121	109	115

	μ	σ
Prior	120	12
Posterior	?	6

MLE solution:

$$\mu_{MLE}^* = \bar{x} = \frac{121 + 109 + 115}{3} = 115 \text{ mmHg}$$

MAP solution:

$$\mu_{MAP}^* = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \cdot \mu_0 = \frac{3 \cdot 12^2}{3 \cdot 12^2 + 6^2} \cdot 115 + \frac{6^2}{3 \cdot 12^2 + 6^2} \cdot 120 = 115.38 \text{ mmHg}$$

Bayesian inference solution:

Posterior distribution:

$$\mu' = \mu_{MAP}^* = 115.38$$

$$\sigma^{2'} = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1} = \left(\frac{1}{12^2} + \frac{3}{6^2} \right)^{-1} = 11.08$$

Posterior predictive:

$$N(x|\mu', \sigma^{2'} + \sigma^2) = N(x|115.38, 11.08 + 6^2)$$